HAIWEN HUANG

Social simulation · Generalization of vision and multimodal models

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POSITIONS

Max Planck Institute for Intelligent Systems, Tübingen, Germany — July 2026 – present
Postdoctoral Researcher

- Group of Prof. Moritz Hardt
- Research on social simulation: what it would mean for an AI simulator to understand society, and how to get closer to that goal

Google, Zurich, Switzerland — July 2025 – Oct. 2025
Research Intern

Megvii Research (Face++), Beijing, China — July 2019 – Sept. 2020
Research Scientist, Face Recognition Team

- Advisor: Xinyu Zhou
- Billion-scale face recognition data annotation; out-of-distribution detection for facial data

EDUCATION

University of Tübingen, Tübingen, Germany — Jan. 2022 – July 2026
PhD in Computer Vision

- Autonomous Vision Group & Bosch Industry-on-Campus Lab; supervisors: Andreas Geiger, Dan Zhang
- Thesis: "Out-of-Distribution Generalization in Visual Perception"
- Defended July 13, 2026

University of Oxford, Oxford, UK — Oct. 2020 – Sept. 2021
MSc in Computer Science, with Distinction

- Dissertation: "On Representation Learning for Deterministic Uncertainty Estimation"; supervisor: Yarin Gal

Peking University, Beijing, China — Sept. 2015 – July 2019
BSc in Computational Mathematics

- Cumulative GPA: 3.73/4.00, Major GPA: 3.80/4.00; advisor: Bin Dong

University of California, Los Angeles, USA — Sept. 2017 – Feb. 2018
Exchange Student, Department of Statistics

- CSC Scholarship recipient; GPA: 4.0/4.0; advisors: Yingnian Wu, Song-Chun Zhu

PUBLICATIONS

1. **Haiwen Huang**, Anpei Chen, Volodymyr Havrylov, Andreas Geiger, Dan Zhang. "LoftUp: Learning a Coordinate-Based Feature Upsampler for Vision Foundation Models." *ICCV 2025.* **Oral presentation.**
2. Volodymyr Havrylov, **Haiwen Huang**, Dan Zhang, Andreas Geiger. "Benchmarking Feature Upsampling Methods for Vision Foundation Models using Interactive Segmentation." *ILR+G Workshop at ICCV 2025.* **Oral presentation.**
3. **Haiwen Huang**, Songyou Peng, Dan Zhang, Andreas Geiger. "Renovating Names in Open-Vocabulary Segmentation Benchmarks." *NeurIPS 2024.*
4. **Haiwen Huang**, Dan Zhang, Andreas Geiger. "Multimodal Dataset Upgrading: a New Challenge for Data Annotation." *ICLR DPFM Workshop 2024.*
5. **Haiwen Huang**, Andreas Geiger, Dan Zhang. "GOOD: Exploring Geometric Cues for Detecting Objects in an Open World." *ICLR 2023.*
6. **Haiwen Huang**, Joost van Amersfoort, Yarin Gal. "Decomposing Representations for Deterministic Uncertainty Estimation." *Bayesian Deep Learning Workshop at NeurIPS 2021.*

7. **Haiwen Huang**, Zhihan Li, Lulu Wang, Sishuo Chen, Xinyu Zhou, Bin Dong. "Feature Space Singularity for Out-of-Distribution Detection." *Safe AI Workshop at AAAI 2021*. **Best Paper Runner-Up.**
8. **Haiwen Huang**, Jiahao Liu, Bin Dong. "Can Convolutional ResNets Approximately Preserve Input Distances? A Frequency Analysis Perspective." *arXiv:2106.02469, 2021*.
9. **Haiwen Huang**, Chang Wang, Bin Dong. "Nostalgic Adam: Weighting More of the Past Gradients When Designing the Adaptive Learning Rate." *IJCAI 2019*.

AWARDS AND HONORS

CVPR Outstanding Reviewer Award	2025
NeurIPS Top Reviewer Award	2024
Best Paper Runner-Up, Safe AI Workshop at AAAI	2021
Outstanding Graduate of Beijing (top 1% in the university)	2019
CSC Scholarship for exchange study at UCLA	2017
Yang Fuqing & Wang Yangyuan Academician Scholarship (top 1% in academic merit and potential)	2016

SERVICE

- Reviewer: NeurIPS, CVPR

TECHNICAL SKILLS

- **Frameworks:** PyTorch, TensorFlow, NumPy, scikit-learn, Hugging Face, Gradio
- **Computer vision:** classification, object detection, (interactive) segmentation, depth/normal estimation, video object segmentation, tracking
- **Foundation models:** DINOv2, CLIP, SigLIP, Mask2Former, Stable Diffusion, Flux; parameter-efficient fine-tuning (LoRA, ControlNet, T2I adapters); LLM APIs for text and multimodal tasks
- **Data-centric evaluation:** data collection, augmentation and synthesis; benchmarking foundation models on VQA and open-vocabulary segmentation